IDX Exchange

Data Analyst Internship

MLS Analytics & Tableau Dashboard Program

Intern Handbook

12-Week Program Guide

Confidential – For Intern Use Only

Welcome to IDX Exchange

Congratulations on joining the IDX Exchange Data Analyst Internship Program. Over the next 12 weeks, you will gain hands-on experience analyzing real MLS transaction data, building market analytics dashboards in Tableau, and communicating meaningful housing market insights.

This handbook is your complete reference guide for the program. Keep it handy throughout the internship – it outlines every week's objectives, the skills you will develop, and what is expected of you at each stage.

Program Mission

Teach students how real estate analytics datasets are produced, cleaned, and transformed into actionable housing market intelligence using industry-standard tools.

Program Overview

The internship is structured as a progressive pipeline – each phase of the program builds directly on the last:

Dataset Exploration	Data Cleaning	Market Analytics	Competitive Intelligence	Dashboard Development	Market Insights

Program Details

Duration	12 Weeks (+ Week 0 Orientation)
Primary Tools	Python (Pandas), Tableau Desktop
Data Source	CoreLogic Trestle API via IDX Exchange pipeline
Dataset Types	Monthly MLS Listing & Sold transaction CSV files
Final Deliverable	Tableau Dashboards + 1-Page Market Intelligence Report + Presentation

Week-by-Week Curriculum

Week 0 – MLS Data Pipeline Orientation

Before analytics begins, you will be introduced to how the MLS datasets used throughout this program are generated. Understanding the data's origins is essential context for everything that follows.

Objective

Understand how monthly MLS datasets are produced from the CoreLogic Trestle API and exported as CSV files for analysis.

Topics Covered

- API authentication and token management
- Pagination handling for large dataset extraction
- Filtering listing and sold records by date ranges
- Exporting monthly datasets to structured CSV format
- Review Trestle Property Metadata (field definitions, data types, and available property attributes)

Example Pipeline Outputs

```
CRMLSListingYYYYMM.csv
CRMLSSoldYYYYMM.csv
```

> **Note**
> Interns may run the extraction script once to observe how datasets are produced. The datasets used throughout Weeks 1–12 are pre-generated and provided at program start.

Required Software Setup

Before your first week begins, please download and install the following tools on your computer.

1. Python IDE (choose one)

You will need a Python coding environment throughout this internship. Download and install one of the following IDEs (Integrated Development Environments) of your choice:

- **Spyder** – Scientific Python development environment, great for data analysis (included with Anaconda). Download at: anaconda.com
- **VS Code (Visual Studio Code)** – Lightweight and widely used editor with excellent Python support. Download at: code.visualstudio.com
- **IDE of your choice** – If you already use a preferred Python environment (e.g., PyCharm, Jupyter Lab), you are welcome to continue using it.

2. Tableau Desktop Public Edition (Free)

Tableau Desktop Public Edition is the free version of Tableau and is the primary visualization tool used in Weeks 10–12 of this program. Download it at **tableau.com/products/public/download** and create a free account.

Weeks 1–2 – Dataset Exploration and Python Setup

You will receive your first real MLS datasets and begin exploring them with Python and Pandas. The goal is to understand the structure and content of the data before any analysis begins.

Datasets Provided

- CRMLSListing202401.csv
- CRMLSListing202402.csv
- CRMLSSold202401.csv
- CRMLSSold202402.csv

Key MLS Fields

`List Price / Close Price`	Transaction pricing data
`Square Footage`	Property size metrics

(LivingArea)	
Bedrooms / Bathrooms	Property configuration
Property Type	Residential, Condo, etc.
Listing Agent / Brokerage	Agent and firm attribution
Days on Market (DaysOnMarket)	Time-to-sell metric

Example Workflow

```
df = pd.read_csv('CRMLSSold202401.csv')

df.head()
df.columns
df.describe()
```

Skills Learned

- Working with CSV datasets in Python
- Pandas DataFrames and data inspection
- Understanding MLS listing vs. sold transaction structure

Weeks 3–4 – Monthly Dataset Aggregation

Individual monthly files are combined into unified datasets that span multiple months, enabling trend analysis over time.

Objective

Load and concatenate multiple monthly MLS files into analysis-ready combined datasets.

Example Workflow

```
sold1 = pd.read_csv('CRMLSSold202401.csv')
sold2 = pd.read_csv('CRMLSSold202402.csv')

sold = pd.concat([sold1, sold2])
```

Outputs

- Combined sold transactions dataset
- Combined listing data dataset

Skills Learned

- Multi-file dataset management
- Data aggregation with Pandas
- Preparing time-series datasets for analysis

Week 5 – Dataset Structuring and Validation

Before analytics begins, the dataset must be inspected and filtered to ensure only relevant residential property records are used.

Tasks

```
# Inspect structure
sold.columns
sold.head()

# Check property categories
sold['PropertyType'].unique()

# Filter residential
sold = sold[sold.PropertyType == 'Residential']

# Validate completeness
sold.isnull().sum()
```

Skills Learned

- Dataset validation and quality checks
- Exploratory data analysis (EDA)
- MLS dataset structure and property type filtering

Weeks 6–7 – Data Cleaning and Preparation

Raw MLS data contains formatting inconsistencies, missing values, and fields that need transformation before analysis. This phase prepares the dataset for reliable analytics.

Tasks

- Convert date fields to datetime format
- Remove unnecessary or redundant columns
- Handle missing values appropriately
- Ensure numeric fields are properly typed

Example

```
sold['CloseDate'] = pd.to_datetime(sold['CloseDate'])
```

Skills Learned

- Data cleaning and transformation
- Preparing analysis-ready datasets
- Working with datetime fields in Pandas

Week 8 – Feature Engineering and Market Metrics

With clean data in hand, you will engineer the key market indicators that power the Tableau dashboards.

Key Metrics to Create

Metric	Formula	Purpose
Price Ratio	`ClosePrice / OriginalListPrice`	Measures negotiation strength
Price Per Sq Ft	`ClosePrice / LivingArea`	Normalizes price across sizes

Days on Market	`DaysOnMarket (raw field)`	Time-to-sell indicator
Year / Month / YrMo	`Derived from CloseDate`	Enables time-series analysis

Skills Learned

- Feature engineering from raw fields
- Housing market analytics concepts
- Time-series variable construction

Week 9 – Outlier Detection and Data Quality

Extreme values in price, square footage, or days on market can distort market averages and trends. You will implement a statistical method to identify and remove these records.

Method: Interquartile Range (IQR)

The IQR method removes records that fall outside a defined statistical range:

```
Q1 = df['ClosePrice'].quantile(0.25)
Q3 = df['ClosePrice'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df['ClosePrice'] >= lower) & (df['ClosePrice'] <= upper)]
```

Why This Matters

Outliers – such as a $50 million estate or a $10,000 distressed sale – can skew median prices and misrepresent the typical market. Removing them produces more reliable and representative analytics.

Skills Learned

- Statistical data cleaning
- Outlier detection using IQR
- Improving dataset reliability for analytics

Weeks 10–11 – Tableau Dashboard Development

With clean, engineered datasets, you will import the data into Tableau and build interactive dashboards that tell the story of the housing market.

Dashboards to Build

Dashboard	Visualizations Included
Market Trends	Median sale price, price per sq ft, sold-to-list ratio, days on market
Market Activity	Homes sold count, new listings volume
Competitive Intelligence	Top agents by volume & units; top brokerages by volume & units
Geographic Analysis	City-level, ZIP code, and county-level market comparisons

Week 12 – Final Presentation and Market Intelligence Report

The final week is dedicated to communicating your findings. You will produce a complete package: polished Tableau dashboards, a concise market intelligence report, and a live presentation.

Deliverables

Deliverable	Description
Tableau Dashboards	All four dashboards fully built and interactive
1-Page Market Intelligence Report	Written summary of key market findings
10–15 Minute Presentation	Live walkthrough of dashboards and insights

1-Page Market Intelligence Report – Structure

- Market Overview – median home price trends, days on market
- Pricing Trends – price per square foot, sold-to-list ratio
- Market Activity – new listings vs. homes sold, transaction volume
- Competitive Landscape – top agents and brokerages by volume and units
- Key Takeaways – 3–5 data-driven insights discovered during analysis

Curriculum at a Glance

Week	Phase / Title	Focus	Skills
0	Pipeline Orientation	Trestle API, data extraction	API basics, data engineering context
1–2	Dataset Exploration	Pandas, CSV inspection	DataFrames, EDA
3–4	Dataset Aggregation	Multi-file concat	Time-series prep
5	Structuring & Validation	PropertyType filter, nulls	Validation, EDA
6–7	Data Cleaning	Dates, types, nulls	Cleaning, transformation
8	Feature Engineering	Price ratio, PPSF, YrMo	Market metrics
9	Outlier Detection	IQR method	Statistical cleaning
10–11	Tableau Dashboards	4 dashboards	Visualization, BI
12	Final Presentation	Report + presentation	Communication, insights

Tools and Technical Setup

Tool	Purpose
Python 3.x	Primary analysis language
Pandas	Data manipulation and aggregation
Jupyter Notebook	Interactive development environment (recommended)

Tableau Desktop	Dashboard building and visualization
Git (optional)	Version control for scripts and notebooks

Python Package Installation
pip install pandas numpy jupyter

Intern Expectations

To succeed in this program, interns are expected to:

- Complete each week's tasks and exercises on schedule
- Ask questions and seek clarification when needed
- Maintain clean, well-commented Python notebooks
- Document findings and observations as you progress
- Participate actively in any check-in meetings or reviews
- Treat the MLS datasets as confidential business data

Reminder
All datasets provided during this internship are sourced from live MLS transaction records. They are confidential and for program use only. Do not share or distribute this data externally.

Glossary of Key Terms

Term	Definition
MLS	Multiple Listing Service – a database of property listings and transactions shared by real estate brokers
CoreLogic Trestle	The API platform used to extract MLS data programmatically
Close Price	The final agreed-upon sale price of a property transaction
Days on Market (DOM)	Number of days between a listing going active and going under contract
Price Per Sq Ft (PPSF)	Close price divided by living area square footage
Sold-to-List Ratio	Close price divided by original list price; above 1.0 indicates a seller's market
IQR	Interquartile Range – the middle 50% of a dataset, used in outlier detection
CRMLS	California Regional MLS – one of the largest MLS systems in the United States

Program Contact and Resources

If you have questions about the program, your datasets, or technical setup, please reach out to your program supervisor at IDX Exchange.

Good luck!

We're excited to have you on the team. This program is your opportunity to build real-world data analytics skills with live market data. Make the most of it, ask questions, and enjoy the process. Welcome aboard.

Appendix A – SQL Cheat Sheet

SQL (Structured Query Language) is one of the most in-demand skills for data analysts. While this program focuses on Python and Tableau, learning SQL alongside your coursework will significantly expand your career options. Use this cheat sheet as a self-study reference.

Basic Query Structure

```
SELECT column1, column2
FROM table_name
WHERE condition
GROUP BY column1
HAVING aggregate_condition
ORDER BY column1 ASC|DESC
LIMIT 100;
```

SELECT Statements

Syntax	Description
`SELECT *`	Select all columns
`SELECT DISTINCT column`	Return unique values only
`SELECT col AS alias`	Rename a column in results
`SELECT TOP 10 / LIMIT 10`	Return only first N rows

Filtering with WHERE

Operator / Syntax	Description
`=, !=, <, >, <=, >=`	Comparison operators
`AND, OR, NOT`	Combine multiple conditions
`BETWEEN 500000 AND 900000`	Filter within a value range (inclusive)
`IN ('Residential', 'Condo')`	Match any value in a list
`LIKE 'Los%'`	Pattern match (% = wildcard)
`IS NULL / IS NOT NULL`	Check for missing values

Aggregate Functions

Function	Description
`COUNT(*)`	Count rows
`SUM(ClosePrice)`	Total of a numeric column
`AVG(ClosePrice)`	Average value
`MIN(ClosePrice)`	Minimum value
`MAX(ClosePrice)`	Maximum value

GROUP BY and HAVING

```
-- Total sales volume and count per city
SELECT City, COUNT(*) AS UnitsSold, AVG(ClosePrice) AS AvgPrice
FROM sold_transactions
WHERE PropertyType = 'Residential'
GROUP BY City
HAVING COUNT(*) > 10
ORDER BY AvgPrice DESC;
```

JOINs – Combining Tables

Join Type	Returns
`INNER JOIN`	Only rows with matches in both tables
`LEFT JOIN`	All rows from left table + matches from right

`RIGHT JOIN`	All rows from right table + matches from left
`FULL OUTER JOIN`	All rows from both tables regardless of match

MLS-Relevant Example Queries

Practice applying SQL directly to the kind of data you work with in this program.

Median sale price by ZIP code

```
SELECT PostalCode, PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY ClosePrice) AS
MedianPrice
FROM sold_transactions
GROUP BY PostalCode
ORDER BY MedianPrice DESC;
```

Top 10 agents by sales volume

```
SELECT ListAgentFullName, COUNT(*) AS UnitsSold, SUM(ClosePrice) AS TotalVolume
FROM sold_transactions
GROUP BY ListAgentFullName
ORDER BY TotalVolume DESC
LIMIT 10;
```

Monthly transaction count trend

```
SELECT YEAR(CloseDate) AS SoldYear, MONTH(CloseDate) AS SoldMonth, COUNT(*) AS
Transactions
FROM sold_transactions
GROUP BY YEAR(CloseDate), MONTH(CloseDate)
ORDER BY SoldYear, SoldMonth;
```

Recommended Learning Resources

Resource	Notes
SQLZoo sqlzoo.net	Free, interactive SQL exercises directly in the browser. Best starting point.
Mode SQL Tutorial mode.com/sql-tutorial	Analyst-focused SQL tutorial covering basic to advanced queries.
W3Schools SQL w3schools.com/sql	Quick syntax reference with live examples for every SQL command.
LeetCode SQL leetcode.com/problemset/database	Practice problems for interview-level SQL. Ideal for career prep.